

SciOdyssey.

A research layer over your agent harness.

PRESERVE THE RESEARCH WORLD · RESET THE RESEARCHER

ROBUST

SUPERVISABLE

BROAD SEARCH

SAME EVIDENCE
FRESH JUDGMENT
A WIDER SEARCH

Research
Runs
Longer.

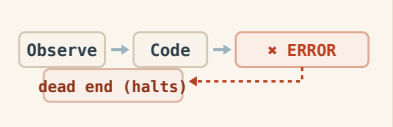
01 / PROBLEM

Why current agents fail.

Sustained autonomous research breaks in three different ways.

01

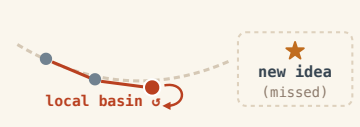
Closed-world fragility
Fixed workflows shatter on unexpected runtime errors.



Unhandled runtime faults halt unassisted runs

02

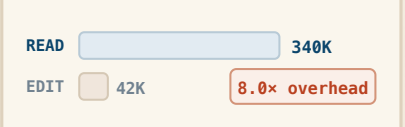
Trajectory momentum
A single context carries past momentum into local basins.



22 runs trapped tweaking parameters in same basin

03

Context inflation cost
Reconstructing context costs far more than the edit itself.



10 files read to change 1 line; burns token budget

DESIGN RESPONSE Keep the world. Reset the trajectory. · Decouple persistent memory from reasoning

- 1 Auto-heal runtime faults
- 2 Fresh reasoning each cycle
- 3 Reusable disk state (no token waste)

02 / SYSTEM

One persistent world. Fresh scientific judgment.

Give the research world and the researcher different lifetimes.

DUAL-LIFETIME LOOP

Persistent world · ephemeral trajectory · audited write-back

INPUTS

TASK

DATA

EVALUATOR

BUDGET

Fresh Scientist
Owns scientific judgment inside one trajectory: hypotheses, code, experiments, interpretation, and pivots.

→

META Review
Audits claims against artifacts, scopes conclusions, and decides what is durable enough to survive.

SELECTIVE PERSISTENCE

Only audited empirical findings, code state, and bounded priors write back to the shared world.

PERSISTENT RESEARCH WORLD

What survives across trajectories

A · EVIDENCE
Knowledge + Ledger
Metrics, failures, reports, and experiment receipts in **ledger.jsonl**.

B · PRIORS
Brief + Hypotheses
Current understanding and exploration leads in **brief.md**, open to revision.

C · STATE
Code + State
Retained implementation, serialized model weights, checks, and git provenance.

CONTEXT RESET

The next Scientist reads verified experience from disk, but does not inherit the previous reasoning trace.

RUNTIME SUBSTRATE

WORKTREE ISOLATION
Independent git worktrees isolate branches and preserve reproducible artifacts.

SELF-HEALING
Traceback → patch → rerun keeps unattended research moving after runtime faults.

PARALLEL SYNTHESIS
Completed branches can be compared post-hoc; useful ideas survive even when a branch loses.

The next Scientist inherits **verified experience** — not prior reasoning momentum.

Decoupled Memory · Audited Persistence · Fresh Scientific Judgment

03 / EVALUATION

Five empirical claims. Measured evidence.

Standardized evaluation on KuaiRand-Pure benchmark · Fixed official evaluator · Complete telemetry accounting

01 MEASURED RESULT

The model got better.
0.6016000 → **0.6059363**
Official FM baseline SciOdyssey submission
+0.0043363 Primary improvement

GAUC **0.6728421**
(+0.0054)

nDCG@5 **0.5390304**
(+0.0033)

LogLoss **0.38102**
(-0.0032)

Feature Engine: 46 categorical cross-interactions + log1p transforms
Architecture: 8-seed Factorization Machine ensemble (NumPy / CPU)

100% FULLY AUTONOMOUS

48.24M LLM tokens

E001-E013 (4 cycles)

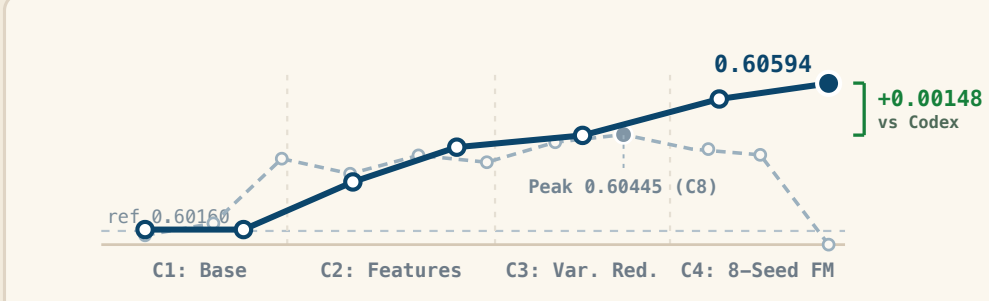
Monotonic gain: 0.6016 → 0.6041 → 0.6059 (100% CPU inference, zero GPUs).

02 SUSTAINED SEARCH

The search stayed productive.

SciOdyssey

Codex Goal



4 cycles

7 Full evaluations

E003-E013 frontier

+0.00148 over Codex

03 SELF-HEALING & GATEKEEPING

The agent recovered from failure.

TIER 1 · RUNTIME SELF-HEALING

CRASH AUTO-RECOVERY

FAILURE

Cycle 1 `numpy.float32` serialization crash

ACTION

Scientist parsed traceback, auto-patched scalar helper

OUTCOME

0 human bugfixes · Unattended rerun succeeded

TIER 2 · META GATEKEEPING

SPURIOUS STATE REJECTION

FALSE CLAIM

Scientist claimed BPR gain on medium proxy

ACTION

META audited `system/data.py`, caught UNK leak

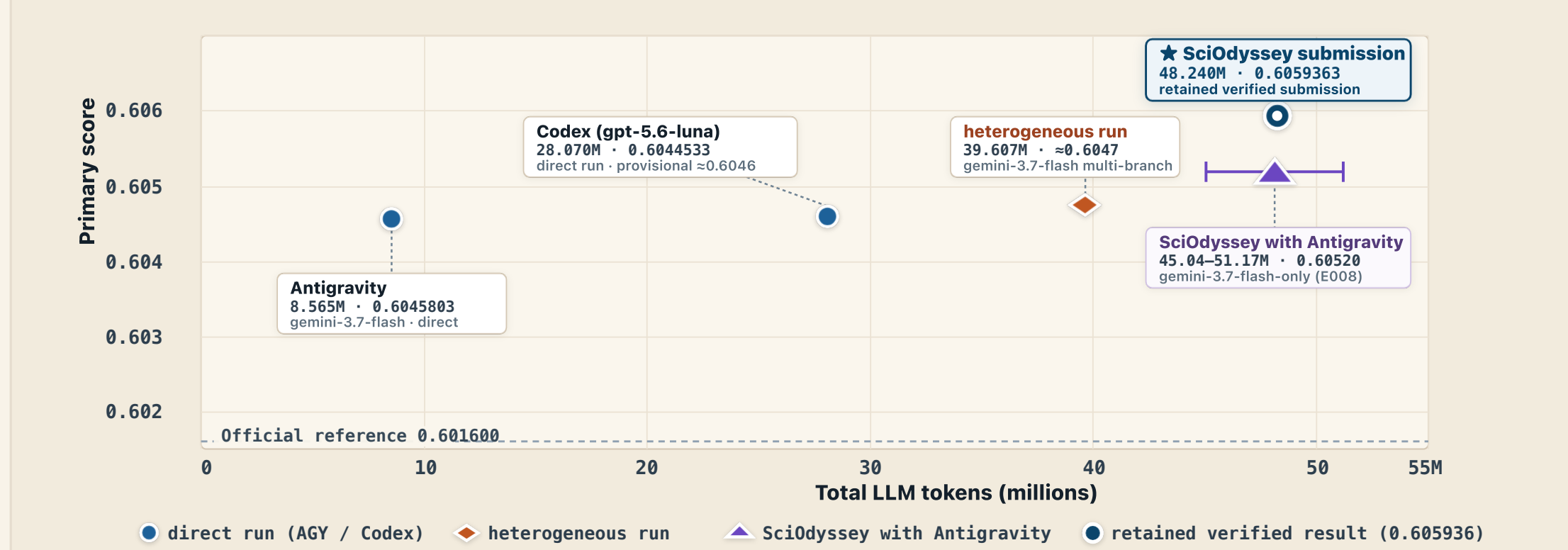
OUTCOME

Promotion blocked · Zero contaminated state

Empirical receipts preserved: runtime self-healing telemetry and independent code-level proxy audits.

05 RESOURCE ACCOUNTING

Score vs token investment.



All results **verified** on KuaiRand-Pure official validation split with **complete reproducible artifacts**.

04 / FINDINGS

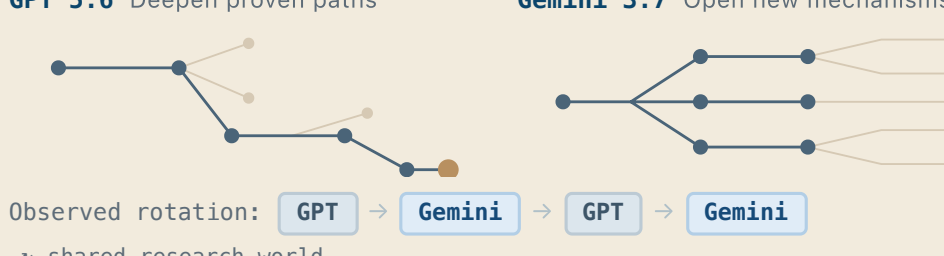
What the run revealed.

01 · MODEL DIVERSITY

Model diversity can help break plateaus.

GPT 5.6 Deepen proven paths

Gemini 3.7 Open new mechanisms



Observed rotation: GPT → Gemini → GPT → Gemini

↳ shared research world

Gemini-only 0.6052 → GPT+Gemini 0.6059

02 · PARALLEL SYNTHESIS

A losing branch can still improve the winner.

A 0.6048 Redundant

B 0.6048 Chosen ✓

C 0.6039 Useful ideas

→ Reviewer Synthesize B+C → Fresh Scientist 0.605536

Selection keeps useful knowledge, not just the top score: Branch C contributed pairwise ranking and rank normalization.

Branch B+C Ideas → Synthesis 0.605536

05 / EXPERIENCE

Start your journey.

task.md

What to solve

Formal task contract: research objectives, dataset inputs, compute constraints, and official evaluation protocol.

Dataset paths & split integrity

Official sandbox evaluator

PERSONAL.md

How to work

User research preferences, model guidance, exploration style, tool permissions, and supervisory checkpoints.

Model rotation (GPT / Gemini)

Supervision & budget thresholds

01 Contract task.md

→

02 Probe step

→

03 Search run

→

04 Branch parallel

→

05 Inspect gui

AUTONOMOUS RUNTIME

Zero manual intervention · Closed-loop agent discovery

1 Unattended Search

Multi-cycle overnight execution; runtime auto-heals unexpected faults without human intervention.

2 Agentic Supervision

Autonomous META agent independently audits empirical artifacts before durable write-back.

3 Parallel Synthesis

Dispatches isolated worktrees; Reviewer agent synthesizes winning mechanisms across branches.

✓ Contract-driven autonomy: Set boundary in task.md; SciOdyssey autonomously iterates, heals, evaluates, and records evidence.

DECLARATIVE PROTOCOL

Autonomous by default.

Supervisable by design.

github.com/zc6600/research-agent-kuairand

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TEAM GOOD4AI · ROBUST · SUPERVISABLE · BROAD SEARCH

A Persistent Research World for Autonomous Science

PROJECT WIKI & DEMO

sciodyyssey.zc6600.wiki

Scan for live report & artifacts